

VibeGuard: 30 cm Ruler (Scale) Measurement Guide

Project: VibeGuard — Edge AI Mechanical Vibration Diagnostic Testbed

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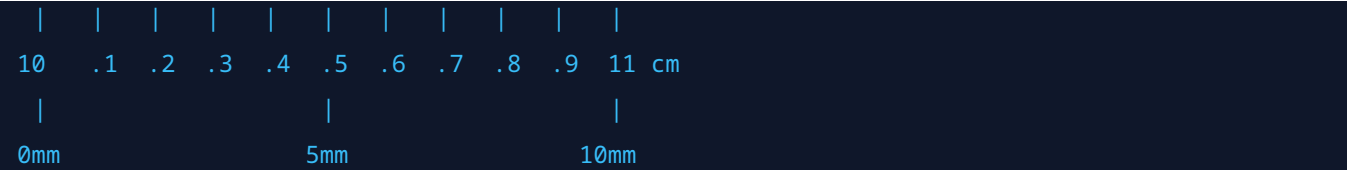
Target Tool: Standard 30 cm / 12-inch Flat School / Engineering Ruler (Metric Scale)

Resolution: 1 cm Major Divisions, 1 mm Minor Divisions (9 small ticks between numbers)

Target Hardware: N20 Micro Metal Gear Motor, Robocraze ADXL345 Breakout (#TJFKQXJUQ), M3 Fasteners, 12 mm Wooden Baseplate

1. How Your 30 cm Ruler Scale is Structured

A standard 30 cm ruler is divided into centimeters (1 cm = 10 mm). Between every centimeter number, there are **9 small tick lines** dividing the centimeter into 10 intervals of **1 mm (0.1 cm)** each:



Quick Decimal & Tick Translation Table

Ruler Reading (Counting from an integer line)	Equivalent Millimeters	Typical VibeGuard Component Matching This Size
0 lines (on the line)	0.0 mm	Starting Reference Mark
1 small tick line	1.0 mm	PCB thickness (1.6 mm $\approx$ 1.5 ticks)
2 small tick lines	2.0 mm	U-bracket metal thickness
2 ticks + halfway to 3rd	2.5 mm	N20 D-Shaft Flat Thickness

Ruler Reading (Counting from an integer line)	Equivalent Millimeters	Typical VibeGuard Component Matching This Size
		(T_{flat})
3 small tick lines	3.0 mm	N20 Shaft Round Diameter (D_{shaft})
3 ticks + tiny hair past	3.2 mm	M3 Clearance Hole Diameter
5 small tick lines (taller middle mark)	5.0 mm	M3 bolt head diameter
7 small tick lines	7.0 mm	Standard M3 washer outer diameter
9 small tick lines	9.0 mm	Wide M3 washer outer diameter
9 ticks + halfway to 10	9.5 mm	N20 usable shaft length
1 full cm + 2 small ticks	12.0 mm	Rotor Arm Pitch & Baseplate Thickness
1 full cm + 5 small ticks	15.0 mm	ADXL345 Mounting Hole Pitch
1 full cm + 7 small ticks	17.0 mm	N20 Motor U-Bracket Hole Pitch
2 full cm marks	20.0 mm	Motor-to-Sensor Separation Gap

2. The 4 Golden Rules for Measuring Small Parts with a Ruler

Measuring tiny mechanical components (2 to 15 mm) with a 30 cm ruler requires special care to avoid large errors:

Rule 1: Never Use the "0" End of the Ruler (The Offset Origin Rule)

Plastic and wooden rulers have worn, rounded, or chipped corners at the "0" end, and many rulers have a blank 2–5 mm dead margin before the zero mark.

Always start measuring from a clean integer line inside the ruler, such as the 10.0 cm or 5.0 cm mark!

Example: To measure a 15 mm hole pitch, align the first hole with the **10.0 cm** mark. The second hole will line up at **11.5 cm** ($11.5 - 10.0 = 1.5 \text{ cm} = 15 \text{ mm}$).

Rule 2: Avoid Parallax Error (Look Strictly 90° from Above)

Because a ruler has thickness (1–2 mm), looking at the tick lines from an angle will shift your reading by up to 1 mm.

- Always look directly perpendicular down onto the ruler line.
- Stand the ruler on its thin edge if possible so the tick marks touch the surface of the component directly.

Rule 3: The "Paper & Pin Puncture" Trick for Hole Spacing

Ruler tick lines cannot reach inside small 3 mm holes. Trying to guess the center of a hole by hovering a ruler above it causes 1–2 mm errors.

Use this foolproof optical transfer method:

- Take a plain white piece of paper or cardboard index card.
- Place the ADXL345 PCB or N20 bracket flat on top of the paper.
- Take a sharp sewing needle, safety pin, or mechanical pencil tip (0.5 mm) and press straight down through the center of both holes to puncture two clean tiny pinpricks into the paper.
- Remove the PCB. Place your 30 cm ruler flat across the two pinholes on the paper.
- Align Pinhole 1 exactly on the **10.0 cm** line. Read the exact tick mark where Pinhole 2 lands!

PCB on Paper:	[Pin 1]	[Pin 2]
	v	v
	(0)	----- (0)
Paper Punctures:	*	*

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Ruler Alignment:      10.0 cm      11.5 cm  --> Exactly 15 mm!

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Rule 4: The "Edge-to-Edge" Method for Holes

If you measure directly on the component:

- Do NOT try to guess where the center of the circle is.
- Instead, align the **left inner edge** of Hole 1 with a full centimeter line (e.g. 10.0 cm).
- Look at the **left inner edge** of Hole 2. Because both holes are identical circles, the distance between their left edges equals the exact center-to-center pitch!

3. Step-by-Step Component Measurement Walkthrough

3.1 N20 Micro Gear Motor Output D-Shaft

```
[ GEARBOX ]
|====|          ===== ROUND SHAFT =====
|====|          |                               |
+---+---+-----+-----+-----+-----+---+
|===== D-FLAT PROFILE =====|
|<----- Usable Length ----->|
```

Step 1: Shaft Round Outer Diameter (D_{shaft})

- Lay the N20 motor flat on a table so the shaft lies horizontally.
- Stand your 30 cm ruler vertically on edge against the shaft tip.
- Align the bottom edge of the cylindrical shaft with the **10.0 cm** line.
- Look across the top of the shaft:

* It will cover exactly **3 small millimeter spaces** (10.3 cm).

* If it covers 3 full millimeter spaces, your shaft is standard **3.0 mm**.

* If it covers noticeably more than 3 ticks (reaching 4 ticks), note it down immediately.

Step 2: Shaft Flat Thickness (T_{flat})

- Rotate the shaft so the flat face is facing directly upwards toward the ceiling.
- Hold the ruler on edge against the end of the shaft.

- Align the curved bottom of the shaft with the **10.0 cm** line.
- Look at the flat top face:

* It should sit **exactly halfway between the 2nd and 3rd tick lines** (10.25 cm = 2.5 mm).

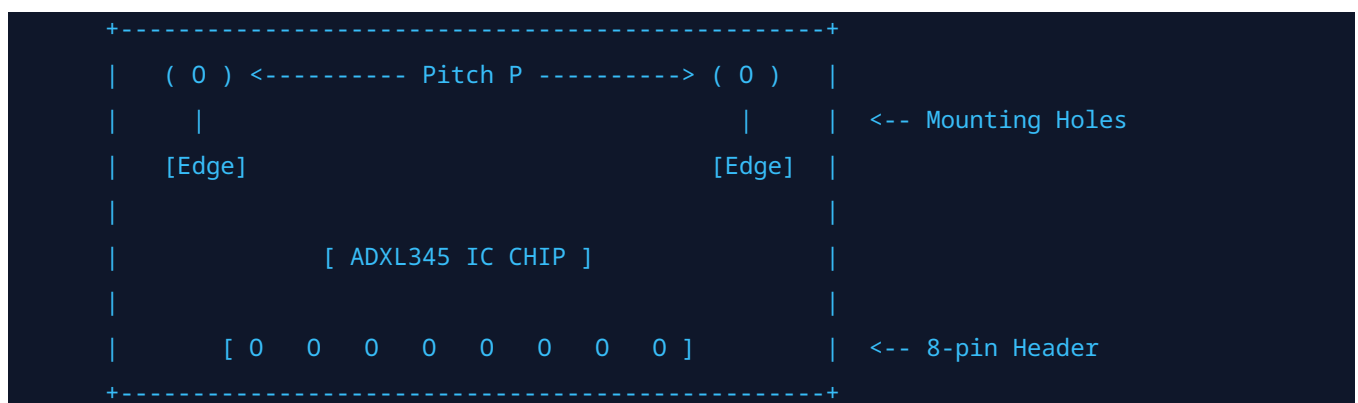
* If it reaches 3 full ticks, there is almost no flat cut! If it only reaches 2 ticks, the flat cut is very deep (2.0 mm).

Step 3: Usable Shaft Length (L_{shaft})

- Lay the motor flat. Place the ruler parallel along the length of the shaft.
- Align the **10.0 cm** line with the front brass collar where the shaft emerges from the gearbox.
- Look at where the steel shaft terminates:

* It should reach 9 small ticks plus halfway to the 10th mark (**10.95 cm**, or approximately **9.5 mm**).

3.2 Robocraze ADXL345 Accelerometer Breakout Board



Step 4: Mounting Hole Pitch (P_{hole})

- **Method:** Use the **Paper & Pin Puncture Trick** (Section 2, Rule 3).
- Place the ADXL345 board on a paper sheet, puncture through both mounting holes with a needle/pin, and measure the distance between the two punctures.
- **Reading on Ruler:**

* Align Pin 1 on **10.0 cm**.

* Pin 2 should land on **11.5 cm** (exactly 1 full cm mark plus the 5th taller millimeter tick mark).

* **Result:** 1.5 cm = 15.0 mm.

(If it lands slightly past 11.5 cm at roughly 11.52 cm, it is standard 0.6 inch / 15.24 mm breadboard grid).*

Step 5: Mounting Hole Diameter ($D_{\text{pcb_hole}}$)

- Lay the ruler flat across the top of the hole.
- The clear hole opening should span just over **3 small tick lines** (approx. 3.2 mm).
- **Quick Test Without Ruler:** Take an M3 machine screw from your hardware kit. Does the threaded shank pass smoothly through the hole without force?

* If **YES:** The hole is ≥ 3.1 mm (confirmed M3 clearance).

* If **NO (binds or won't enter):** The hole is 2.5 mm or 2.6 mm (M2.5 size).

Step 6: Backside Solder Blobs / Header Pins

- Turn the board sideways so you can see the back profile.
- Hold the ruler against the back fiberglass plane.
- Inspect how far the solder joints or clipped pin stubs stick out:

* Typically **1 small tick to 1.5 ticks** (1.0–1.5 mm).

3.3 N20 Aluminum U-Bracket

Step 7: U-Bracket Mounting Hole Pitch (P_{bracket})

- Use the **Paper & Pin Puncture Trick:** Place the aluminum bracket flat on paper, poke a pin through both bottom mounting holes.
 - Align Pin 1 on **10.0 cm**.
 - Pin 2 should land on **11.7 cm** (1 full cm plus 7 small ticks).
 - **Result:** 1.7 cm = 17.0 mm.
-

3.4 M3 Hardware & Baseplate

Step 8: Steel Washer Outer Diameter (D_{washer})

- Place a washer flat on a white sheet.
- Align the left edge of the washer with **10.0 cm**.
- Look at the right edge:

* If it reaches **7 small ticks** (10.7 cm), you have a standard 7.0 mm DIN 125 washer.

* If it reaches **9 small ticks** (10.9 cm), you have a wide 9.0 mm DIN 9021 washer.

(Both work! Our 15 mm riser block gives +7.5 mm clearance for 7 mm washers, and +6.5 mm clearance for 9 mm washers).*

Step 9: Wooden Baseplate Thickness (T_{plank})

- Hold the ruler against the outer cut edge of the wooden board.
- Align the bottom face with a full centimeter line (e.g. 5.0 cm).
- Count the ticks to the top face:

* Should be 1 full cm plus 2 small ticks (**1.2 cm = 12.0 mm**).

4. 30 cm Ruler Audit Sheet (Fill & Send Back)

Align each feature with the **10.0 cm** reference mark on your 30 cm ruler, record the second mark you see, and fill in the table below:

#	Component & Feature	Start Mark on Ruler	End Mark on Ruler	Your Ruler Reading (cm + ticks)	Converted mm	Expected Nominal
1	N20 Shaft Round OD	10.0 cm	10._ cm	10 cm + __ ticks	__._ mm	3.0 mm (3 ticks)
2	N20 Shaft Flat Thickness	10.0 cm	10._ cm	10 cm + __ ticks	__._ mm	2.5 mm (2.5 ticks)
3	N20 Shaft Usable Length	10.0 cm	10._ cm	10 cm + __ ticks	__._ mm	9.5 mm (9.5 ticks)
4	N20 Bracket	10.0 cm	11._ cm	11 cm + __	__._ mm	17.0 mm

#	Component & Feature	Start Mark on Ruler	End Mark on Ruler	Your Ruler Reading (cm + ticks)	Converted mm	Expected Nominal
	Hole Pitch			ticks		(1.7 cm)
5	ADXL345 PCB Hole Pitch	10.0 cm	11._ cm	11 cm + __ ticks	__._ mm	15.0 mm (1.5 cm)
6	ADXL345 PCB Hole Size	M3 bolt test: Does an M3 screw slide smoothly through the hole?	☐ YES ☐ NO	3.2 mm		
7	ADXL345 Solder Protrusion	10.0 cm	10._ cm	10 cm + __ ticks	__._ mm	1.0–1.5 mm
8	M3 Washer Outer Diameter	10.0 cm	10._ cm	10 cm + __ ticks	__._ mm	7.0 or 9.0 mm
9	Wood Baseplate Thickness	10.0 cm	11._ cm	11 cm + __ ticks	__._ mm	12.0 mm (1.2 cm)